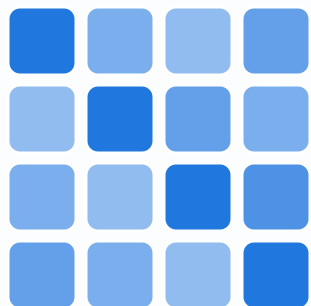


Transformer families differ by allowed information flow

The core block is similar; the mask and cross-attention wiring determine whether the model embeds, generates, or maps one sequence to another.

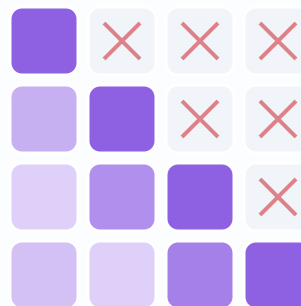
Encoder-only



bidirectional mask

BERT-style
classification, embeddings
retrieval

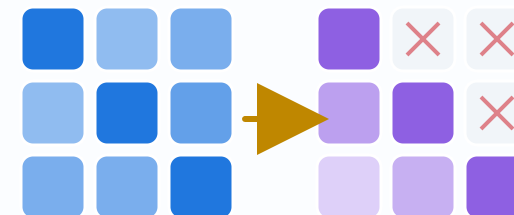
Decoder-only



causal lower triangle

GPT-style
generation, chat
code, agents

Encoder-decoder



encoder memory + causal decoder

T5 / BART-style
translation
summarization